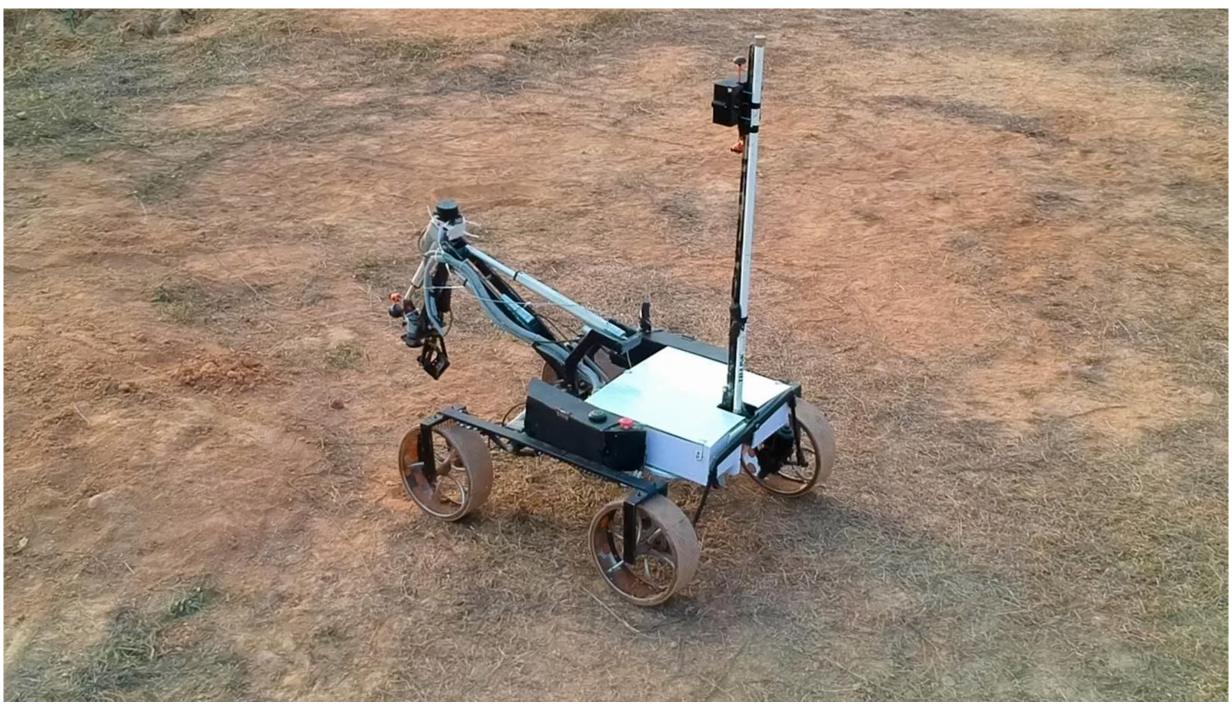
System Acceptance Review

BRACU Mongol-Tori

BRAC University

University Rover Challenge – 2020



Team Lead: Md. Firoz Wadud
Email: mdfirozwadud.adm@gmail.com

Optional Contact:
Name: Raihan Rahman
Email: raihan.rahman1010@gmail.com

1. Introduction: BRACU Mongol Tori is a team led by robotics enthusiasts from different departments of BRAC University mostly consisting of engineering students. BRAC University's Robotic Club (ROBU) serving as a parent organization for financial and mechanical engineering support. Our rover is capable enough to assist astronauts as well as perform science-related research tasks. In the last few years, the team has already made some remarkable changes in the overall structure of the rover. Currently, the major updates include the development of the autonomous systems, electronics, and science setup. Along with that, minor altercations like modifying chassis, improving arm and claw, communication system, control GUI can also be observed. In this report, analysis has been made on the Core Rover Systems, Approach to Competition Tasks, Testing and Operations, Team Budget, Gantt Chart, and Science Task according to the specifications of URC 2020.

2. Core Rover Systems: Based on our observations on URC 2019 and after several trials and errors we have decided to bring in some subtle changes in the mechanical setup and some massive changes in the electronics and autonomous and science segment.

2.1 Mechanical: A strong arm equipped with a worm gear at its base for having the freedom of 360-degree rotation has been developed for this year, along with that, we are working on a new cartesian arm which is under development. Our four-wheeler setup produced satisfactory results for us last time, hence we are retaining this setup.

a **b** **c** **d**

Figure: 1

2.1.1 Chassis and Suspension: We have developed an H shape chassis which is 0.7-meter-long and 0.55-meter-wide to hold the circuit box and arm. This shape of our rover is quite unique which is stiff and firm. Moreover, in the chassis, triangulation has been installed in order to reduce shape distortion while traversing rocky terrain. The heavy space frame transmits the flexing load as tension and compression loads along the length of each strut. Besides that, our rover has been incorporated with a modified version of the rocker-bogie suspension system with four-wheelers. Our system consists of two bogies directly connected to the chassis and also it is supported by U shaped differential bar at the rear part of the system. In the differential bar, we have used universal joints for maximizing the rotational angle of the bogies. The unique feature in this system is it has the ability to expand the bogies. Here we have used two heavy-duty feedback actuators and the length bogies can turn to 1.2m from 0.76m. During the vertical drops and slopes, the extended suspension provides more durability. Figure 1(a) shows our full working rover's design. Whereas, figure 1(d) shows the design of our extender which adds a unique feature to our rover by increasing the length of its base and preventing the rover from flipping over during vertical drop.

2.1.2 Wheel: The previous version of the wheel has been retained as it proved quite efficient for us in the last run. For URC 2020 our wheel has 0.3-meter diameter and 0.10-meter width. It has been developed with a stainless steel ring through U shaped rim. In order to make our wheel lightweight and having a high breaking point, it has been made of stainless steel. This design makes the rover better than the previous one by having the shock-absorbing capability. By drilling small holes we have made the wheels lighter. Apart from this, we used rubber pad on the wheel for rectifying 360-degree rotation, climbing rock and dealing with a vertical drop.

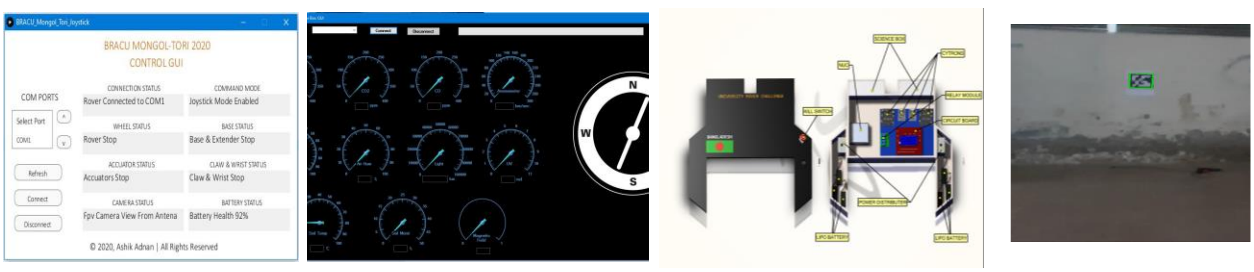
2.1.3 Arm: Our rover has a strong robotic arm that has six-degrees of freedom (base, shoulder, elbow, wrist) and two degrees of freedom in the end effector. Figure 1(b) shows that the arm consists of four feedback actuators and two dc motors. At the top of the arm, the gear mechanism has been integrated because it gives the end effector more degrees of rotational freedom without wire twisting. In the claw, a feedback actuator system gives superior gripping flexibility. On the other hand, we are using a worm gear in the base for easy and precise rotation. Using worm gear gives us an extra feature, if the motor fails for any reason the gear will be locked and prevent further rotation. The claw that has been used is lightweight, strong and stable because it is suited for different tasks. We have concentrated on strength and stability for the accurate movement of the arm. Furthermore, there are 3 feedback end effectors that have been included with the arm to increase our precision while performing different tasks. A cartesian arm is being worked on as shown in fig (c) which is under development.

2.2 Communications: For a seamless flow of data to and from the rover, we have configured a peer-to-peer network model using two high-end 2.4GHz routers. In the base station, Ubiquiti Rocket M2 paired with a 15dBi directional sector antenna(ANT232D15T-120DB) is used. The sector antenna is connected to a height-adjustable stand and paired with a rotator (G-5500) which is controlled using an elevation azimuth controller that allows the antenna to be rotated 360 degrees. As a result, the sector antenna has a 360-degree horizontal beam pattern. On the other hand, the rover is equipped with a Ubiquiti Bullet M2 fitted with a 12dBi omnidirectional antenna (TL-ANT2412D) for 360 degrees signal coverage. The omnidirectional antenna enables the rover to be controlled no matter the orientation. The rover is fitted with an Intel NUC, which connects to the Bullet M2, enabling data connectivity between the base station and the rover. The network is configured with a static IP address configuration to avoid IP address entanglement. The 2.4GHz peer-to-peer network enables smooth control of the rover for over 1 kilometer. For vision, both 2.4GHz and 5.8GHz bands are used. Two IP cameras installed in key points of the rover. The IP cameras are accessed through the 2.4GHz network. Since with increasing range the 2.4GHz system's latency increases, 5.8GHz FPV system is also used. Four FPV cameras with 2.5mm lens are placed around the rover for a balanced field of view. The FPV system is equipped with two 800mW transmitters and two receivers which are connected with polarized 3dbi mushroom antennas, thus the FPV system gives excellent and reliable video feed even after exceeding the maximum range for 2.4GHz system. At a time two of the four FPV cameras can be accessed while the others can be accessed via an FPV-switcher from the control system. Lastly, a precise DC-DC power supply is built to meet various power requirements of the communication system.

2.3 Control system: A portable base station has been built that will be operated with 1 laptop and 3 monitors for camera view as well as a joystick, gamepad and a keyboard for smooth controlling of the rover.

2.3.1 Electronics: The prime upgrade that we are introducing this year is the ease of installation of the circuit boards which helps to debug the circuit easily. An atmega2560 is employed for processing all the commands. Back emf protected 30A motor drivers have been used for smooth PWM control of the wheel. Along with this, 13A motor drivers and 10A relay modules have been used for precise maneuvering of the arm. Notably, we have developed a reverse polarity protection circuit on the power distribution board by using Schottky diode which has the lowest voltage drop to save the devices from unexpected human errors. Unlike previous years where we struggled with equal power consumption, this time a centralized power system has been implemented. It helps by providing an equal run time for the whole system. We are using six 10,000mAh batteries which powers up the whole system through the power distribution circuit. This power unit has an average run time of 45 minutes. Lastly, a customized high ampere kill switch has been installed for the emergency shutdown of the rover.

2.3.2 Software: A new Graphical User Interface has been built by the software and control team for control, science, and mapping with an advantageous programming framework. The mapping GUI tracks the rover continuously on a pre-loaded offline map using the latitude and longitude received from the GPS sensor. We have built up the control GUI shown in Figure 2(a) using JAVA programming language. This framework follows the Transmission Control Protocol (TCP/IP) where the base station acts as the client while the rover acts as the server. One of the indispensable upgrades that we have continued from last year is the feedback system. We have implemented feedback receiving environment in our control GUI in order to use feedback end effectors along with feedback mechanisms. In addition, we have developed the science GUI using Java which can demystify the sensor readings into custom measures and offers graphical patterns on the screen making it simple to recognize distinctive conditions of observations. We have made a control program shown in Figure 2(b) for an onboard science test in our science GUI.



a b c d

Figure 2

3. Approach to Competition Missions: In order to cover all the tasks to be performed at URC we arranged mock competition practice on locations that exhibit similarities with that of URC. All the tasks were performed in a certain allocated time.

3.1. Extreme Retrieval and Delivery Mission:

The modified rocker-bogie suspension and upgraded chassis are suitable for performing Extreme Retrieval and Delivery Mission. The rover is designed to move easily up to one kilometer. While moving through any rough terrain, its suspension system can withstand sudden shocks. Our custom-made wheels are capable of having superior grips over rocky and sandy areas. During the vertical drop, the expanded rover body allows the rover to land firmly. Its strong-arm can lift heavyweight objects like toolboxes or water bottles easily.

3.2. Equipment Servicing Mission:

For equipment servicing, we have a separate two-finger claw for a precise switch and keyboard operation that can be rotated as well. In addition, a feedback end effector has been added for precise movements and angle measurements. For this, the rover has been modified with angle calculation instead of using a camera to detect movements. The formula is ?? = ??c??? (?^2+?^2−?1^2)/ 2??. In this way, the necessary movements can be determined along with the ones that have already occurred.

3.3. Autonomous Traversal Mission:

From our previous experiences, instead of trying to develop a system from scratch, we are capitalizing on an existing system with some new approach. We have employed Pixhawk 4.0 with ardurover firmware uploaded in it and the Radiolink SE100 GPS-Compass module for seamless autonomous traversal to the waypoint. This system has 50cm positioning accuracy. We are using “Mission planner” as the base station user interface. We can easily set commands like “waypoint” for traversing along with the waypoints, “hold” for holding the rover at a location. We can also create a search pattern with the help of the loiter mode. We are applying the skid steering system on pixhawk which generates two individual output values for the left and right wheel motor. These values are then received by an ATmega2560 which maps these values and sends appropriate commands to the motor drivers. Additionally, it decides when to blink the specific LEDs. For deciphering the AR tags we are using image processing with the help of OpenCV as shown in fig 2(d). We do the detection task by contour detection and homography. This system has an accuracy of 80%. For obstacle detection, we are employing a Rplidar that rotates 360 degrees and continuously fires laser pulses at around 150,000 pulses per second and detects the object while receiving the emitted pulses giving an accurate object detection. If an object is detected then it sends a signal to the ATmega2560 to avoid the object and keep following the waypoints.

4. Safety measures: According to the rules and also concerning the safety of the rover, we have developed a “kill switch” which will immediately force shutdown the rover mechanisms when needed. Additionally, to ensure the safety of our main circuit board we have implemented an overcurrent protection system.

5. Testing and Operations: Team Mongol Tori conducted various examinations in three different steps to ensure the proper functionality of the rover.

a. Sub-System Testing:

The team tested all the systems in solid works before implementing the design, and those accordingly from 18/10/2019 to 10/12/2020. The subsystems include wheels, suspension, and chassis, end effector, sample collector, on-board science test and software. The testing of the rover wheels was done by a custom ramp on 27/11/2019. After that, on 22/11/2019, an indoor test was conducted on the rover chassis and suspension. Considering various types of gear mechanisms, the worm gear assembly was chosen for the end effector. This module was tested on 08/12/2019 using small objects like a screwdriver, hammer, etc to grab. Finally, a platform-independent control GUI which is compatible with any system was developed. With this advantage, the control GUI was tested on a prototype rover on 28/11/2019.

b. Complete System Testing:

The complete configuration on the rover was essentially tested to get a positive confirmation with the new updated communication and electronics system in January 2020. Fortunately, the communications and electronic system had a good performance in URC’2019. Moreover, we have participated in the Indian Rover Challenge (IRC)-2020 from 17/01/2020 to 19/01/2020 with these systems which worked accordingly. Hence, the same systems were built upon again. On 04/02/2020, the total robotic arm system and end effectors were tested by picking up various objects and collecting different soil samples.

c. Operational Testing:

Our four main tasks of URC’2019 acted as the base for operational testing. A remote place, identical to the main competition ground, consisting of a vertical drop point, steep slope and different types of rough terrain was selected to conduct the testing. The team went to this test site for extreme retrieval tasks with the rover and began the tests from December 2019. The testing of equipment servicing was done under a dummy set up on 17/12/2020. This included turning a switch on-off, turning knobs, collecting cache boxes and undoing latch, etc. A location with GPS availability was chosen for the autonomous test on 02/01/2020 and the tests were successfully conducted. Some soil samples were collected by the team using the rover’s sample collector to perform the science task and on-board tests in the science module part. About the science module, the collected soil samples were tested on 18/02/2020 for signs of life and microbial biomass testing, capillary water flow testing and amino acid testing were successfully carried out.

Gantt Chart:

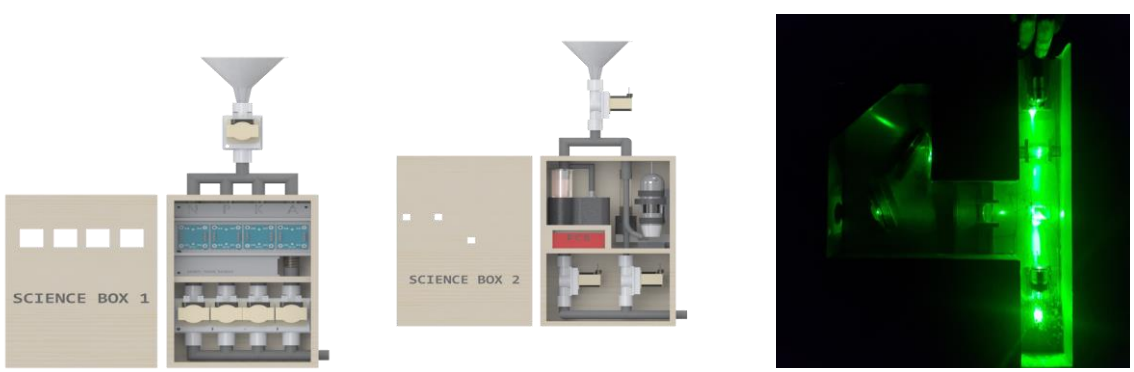
Project Starting Date	Project Finishing Date	Duration
September 06, 2020	July 31, 2020	11 Months

Task	Sep '19	Oct	Nov	Dec	Jan '20	Feb	Mar	Apr	May	Jun	Jul
Team Recruitment (Sep 10,19 to Oct 24,19)											
Sponsor Collection, Project Plan (Oct 10,19 to Apr 25,20)											
Development Phase:											
3D Model & Designing (Oct 18,19 to Dec 10,19)											
Chassis & Suspension Design Implementation (Nov 22,19 to Dec 05,19)											
Custom Ramp and Wheel Design (Nov 27,19 to Dec 01,19)											
Rover Arm(Dec 08,19 to Jan 12,19)											
Control Circuit & Software (Nov 28,19 to Dec 12,19)											
Command Station (Jan 11,20 to Feb 13,20)											
Testing Phase:											
Wheel & Suspension Test (Nov 29,19 to Dec 07,19)											
Extreme Retrieval Test (Dec 02,19 to Jan 31,20)											
Equipment Servicing Task (Dec 17,19 to Feb 11,20)											
Autonomous Test (Jan 02,20 to Mar 28,20)											
Science Test (Feb 04,20 to Apr 23,20)											
Submission of SAR (Feb 28,20)											
Preparation for URC-2020 (if SAR approved) (Apr 01,20 to May 20,20)											
Future Rover Development Plans (From Jun 14,20)											
Promotional Activities (Sep 27,19 to Jul 30,20)											

Team Budget:

Income	Amount	Expenses	Amount
Initial Budget	$480.00	New Components	$3543.00
Budget from University	$2,960.00	Promotional Activities for Fund Raising	$515.00
Internal Crowd-Funding	$2,214.00	**Development Cost**	
Amount from Sponsors:		Mechanical	$635.00
NasterLab	$155.00	Communication	$484.00
Guardian Life Insurance	$715.00	Electronics and Battery	$317.00
Committed Sponsors (If SAR Approved):		Science Module and Setup	$610.00
BRAC University (Plane fare)	$11,810.00	Control and Processing Devices	$745.00
US-BANGLA Airlines	$6500.00	USA Travel Cost (10 person)	$17,717.00
Infinity	$388.00	**Surplus**	**$6,339.00**
Walton	$5683.00		
Total Income	**$30,905.00**	**Total Expense**	**$24,566.00**

Science Task:
This year we have a change in the whole science setup, though most of the analysis remains the same, we are introducing Raman spectroscopy for measuring the properties of soil samples more accurately. Unlike the previous year, this year we have made our science setup capable of testing samples from multiple sites and have an enhanced vision on the claw for rock examination. Also, we have designed our science task to take a uniform distribution of soil samples for each of the distinct tests. We have divided our science task into three different segments, 1. Digging module and sample collector, 2. Environmental data collecting sensor box, 3. A science box as an automatic onboard laboratory. We have an arm with six degrees of freedom, equipped with an excavator claw that can dig as deep as 20 cm or more through any compact soil surface and dug out the soil and put it in the sample box. In a single grab, the claw can extract around 80gm of soil. The on-board sensor box is equipped with Air Temperature Sensor (DHT22), Air Humidity Sensor (DHT22), Soil Temperature Sensor(DS18B20), Soil Moisture Sensor (DS18B20), CO Sensor (MQ7), CO2 Sensor (MQ135), H2 Sensor (MQ8), Light Intensity Measurement Sensor (LDR), UV Light & radiation Measurement Sensor (ML8511), Hall Effect Sensor for magnetic field detection, Compass Sensor for Navigation, Barometric pressure sensor 360-degree Panorama View of Arena, pH Soil Probe Sensor, and a endoscopic camera capable of taking clear pictures of the trench. From the results of these sensors, we can get a decent idea of planetary environmental data, climate, soil composition and formation of the landscape. Among the sensors, soil temperature, soil moisture and pH probe are connected at the top of the arm. For automatic onboard laboratory tests, we are taking soil samples and analyzing them to find any traces of either extinct or extant life form and water history. To complete these tasks our onboard laboratory is divided into two segments, in first segment we have performed the Biomass test and Water Flow Capillary test. For the Biomass test, a load cell is used to detect weight variation and a nichrome wire heater for heat generation. The formula is: **Biomass = (difference of weight/previous weight) * 100**. Moreover, a centrifugal tube is used in Water Flow Capillary test to hold the soil & a servo motor pushes a syringe that can provide water in the tube. It provides the water absorption rate of soil in that area. This test indicates whether the soil surface is able to hold the water on the surface as surface water or absorb and store it as groundwater. A sample collecting chamber is incorporated to store the pure form of sample for future experiments. The second segment is assigned with the task of carrying out a quantitative analysis of the sample using the spectroscopic method, a light source coupled with a sensor is used to measure the absorbance of light. Using this method, we can find the amount of Nitrogen, Potassium, and Phosphorous which are vital and important elements of soil. For Amino Acid test along with a nichrome wire heat generator and test tube, a ninhydrin reagent is used to detect the presence of amino acid in the sample which plays a key role in the sustenance of life. To analyze the microscopic properties of rocks and soil we are using a digital microscope.



(a) (b) (c)

Figure 3